

NLU course project

Name Surname (mat. number)

University of Trento

name.surname@studenti.unitn.it

1. Introduction

The entire workflow was organized as follows: model definitions were first established, followed by a training-validation phase used for hyperparameter tuning. Finally, for each section, the model with the best-performing hyperparameter set was evaluated on a separate test set.

2. Implementation details

The implementation builds on the original Lab5 codebase and was adapted to meet project goals. Each model has a unique identifier based on its architecture and hyperparameters, with a dedicated directory under `models/` for storing related data. In Part 2A, each model used a different training/dev split, requiring a separate vocabulary stored as `vocabulary.json` in its directory. In contrast, Part 2B used a shared data split for all models, reducing variability in the experimental setup, isolating the effects of architectural differences and weight initialization as the primary variables between models. Training metadata and training log were saved in `training_data.json` and `training.txt`. Models were evaluated regularly—every 5 epochs in Part 2A (200 epochs total), and every epoch in Part 2B (40 epochs). Early stopping was applied if no improvement occurred over three evaluations. The best-performing model was retained along with its corresponding evaluation results (`dev_data.json`). Validation performance metrics are logged in `dev.csv` and they also include 95% CI for both slot F1 and intent accuracy. Following training, the best model is evaluated on the test set. The final test results are recorded in both `test_data.json` and `test.csv`.

3. Results

3.1. PART 2A

To establish a reliable baseline, the IAS model defined in Lab 5 was trained using different Learning Rates: 0.0001, 0.001, 0.01 and Batch Sizes: 128, 64, 32. The best-performing configuration was achieved with a LR of 0.001 and a BS of 128. Based on this result, I experimented with increasing dimensions to 400-600 and 600-900 to evaluate whether larger hidden and embedding sizes could improve performance.

Simple IAS							
#L	LR	BS	Hid	Emb	DO	Slot F1	Intent Acc
1	1e-04	128	200	300	/	0.9668	0.9779
1	1e-03	128	200	300	/	0.9699	0.9779
1	0.01	128	200	300	/	0.9675	0.9699
1	1e-04	64	200	300	/	0.9654	0.9799
1	1e-03	64	200	300	/	0.9640	0.9779
1	0.01	64	200	300	/	0.9598	0.9679

#L	LR	BS	Hid	Emb	DO	Slot F1	Intent Acc
1	1e-04	32	200	300	/	0.9678	0.9779
1	1e-03	32	200	300	/	0.9698	0.9759
1	0.01	32	200	300	/	0.9499	0.9598
1	1e-03	128	400	600	/	0.9689	0.9739
1	1e-03	128	600	900	/	0.9699	0.9799

Table 1: Results of SimpleIAS.

BIDIRECTIONAL LSTM: Next, bidirectionality was added to the IAS model. Using the same Learning Rate (0.001) and Batch Size (128), I evaluated performance with different model dimensions: 200-300, 400-600, and 600-900. This modification led to clear improvements in both Slot F1 score and Intent Accuracy.

B-IAS							
#L	LR	BS	Hid	Emb	DO	Slot F1	Intent Acc
1	1e-03	128	200	300	/	0.9796	0.9819
1	1e-03	128	400	600	/	0.9796	0.9799
1	1e-03	128	600	900	/	0.9808	0.9859

Table 2: Results of Bidirectional IAS.

DROPOUT: In the final step, I added a dropout layer after the LSTM to mitigate overfitting. Starting from the best previous configuration LR: 0.001, BS: 128, dimensions: 600-900, I tested different dropout values: 0.3, 0.5, 0.7. The best performance was achieved with a Dropout of 0.5. I then evaluated the effect of varying the Batch Size. Finally, I increased the model capacity by stacking multiple LSTM layers. As a reference point, I also tested the configuration proposed by Liu and Lane [1].

B-IAS with Dropout							
#L	LR	BS	Hid	Emb	DO	Slot F1	Intent Acc
1	1e-03	128	600	900	0.3	0.9790	0.9819
1	1e-03	128	600	900	0.5	0.9787	0.9839
1	1e-03	128	600	900	0.7	0.9793	0.9819
1	1e-03	64	600	900	0.5	0.9816	0.9839
1	1e-03	32	600	900	0.5	0.9820	0.9799
1	1e-03	16	600	900	0.5	0.9802	0.9839
2	1e-03	32	600	900	0.5	0.9814	0.9859
3	1e-03	32	600	900	0.5	0.9744	0.9859
4	1e-03	32	600	900	0.5	0.9778	0.9779
2	1e-03	32	600	900	0.3	0.9805	0.9859
2	1e-03	32	600	900	0.7	0.9790	0.9839
1	1e-03	16	128	128	0.5	0.9776	0.9839

Table 3: Results of Bidirectional IAS with Dropouts.

After the training and tuning phase, the best model from each section was evaluated on the test set. Results show a better behavior for the Dropout solution, but it is important to see that it has bigger dimensions.

Test Result							
#L	LR	BS	Hid	Emb	DO	Slot F1	Intent Acc
Simple IAS							
1	1e-03	128	200	300	/	0.9253	0.9272
B-IAS							
1	1e-03	128	600	900	/	0.9480	0.9440
B-IAS with Dropout							
2	1e-03	32	600	900	0.5	0.9526	0.9619

Table 4: Results of 2A Testing

An interesting observation is that, while the Bidirectional and Dropout models achieved similar intent accuracy on the development set, the Dropout model performed better on the test set, indicating stronger generalization capabilities.

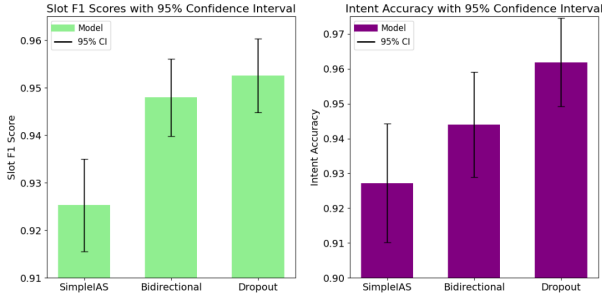


Figure 1: Intents and Slots Error Bars for the 3 tested models

3.2. PART 2B

BASE BERT: In the BERT-based experiments, the initial focus was on the BERT base model. The first set of tests used a the same hyperparameters of the reference paper [2]: Dropout Rate of 0.1, a Batch Size of 128, and different Learning Rates: 1e-5, 5e-5, 1e-4, 5e-4. Results showed that high learning rates, such as 5e-4, led to a significant drop in performance, while lower values produced more stable and accurate models. After selecting 1e-4 as a suitable learning rate, further experiments varied the Batch Size: 32, 64, 128 and Dropout: 0.05, 0.3, 0.5 to examine the model’s sensitivity to regularization.

Once the best configurations for the base model were identified, the focus shifted to BERT large, which required smaller Batch Sizes: 16, 32, 64 due to increased computational demands. Learning rates of 5e-5 and 1e-5 were tested in combination with dropout values 0.05, 0.1 and 0.15. As with the base model, experiments followed a systematic approach to analyze the effect of each parameter. At the end since the Intent Task get always better result I try to give more relevance to the slot task in the training with a weighted loss:

$$\text{loss} = \alpha \cdot \text{loss}_{\text{intent}} + (1 - \alpha) \cdot \text{loss}_{\text{slot}}$$

The alpha value tested were 0.4, 0.3 and 0.2.

BERT Base				
LR	BS	DO	Slot F1	Intent Acc
1e-05	128	0.1	0.9773	0.9900
5e-05	128	0.1	0.9832	0.9900
1e-04	128	0.1	0.9826	0.9920
5e-04	128	0.1	0.9541	0.9719
1e-04	64	0.1	0.9841	0.9940
1e-04	32	0.1	0.9829	0.9900
1e-04	128	0.2	0.9855	0.9900
1e-04	64	0.2	0.9832	0.9900
1e-04	32	0.2	0.9835	0.9920
1e-04	128	0.3	0.9826	0.9900
1e-04	64	0.3	0.9829	0.9900
1e-04	32	0.3	0.9826	0.9900
1e-04	128	0.05	0.9849	0.9880
1e-04	64	0.05	0.9852	0.9839
1e-04	32	0.05	0.982	0.9839

Table 5: Results of BERT base models.

BERT Large					
LR	BS	DO	α	Slot F1	Intent Acc
1e-05	16	0.1	/	0.9861	0.9900
1e-05	32	0.1	/	0.9858	0.9900
1e-05	64	0.1	/	0.9858	0.9960
1e-05	16	0.15	/	0.9849	0.9900
1e-05	32	0.15	/	0.9858	0.9960
1e-05	64	0.15	/	0.9841	0.9900
1e-05	16	0.05	/	0.9867	0.9940
1e-05	32	0.05	/	0.9832	0.9960
1e-05	64	0.05	/	0.9852	0.9920
1e-05	64	0.1	0.4	0.9852	0.9900
1e-05	64	0.1	0.3	0.9882	0.9940
1e-05	64	0.1	0.2	0.9849	0.9940

Table 6: Results of BERT large models.

Test Result					
LR	BS	DO	α	Slot F1	Intent Acc
BERT Base					
1e-04	128	0.2	/	0.9581	0.9765
BERT Large					
1e-05	64	0.1	0.3	0.9555	0.9765

Table 7: Results of 2B Testing

From the test results, it can be observed that both BERT base and BERT large models perform similarly. However, BERT base is preferable due to its significantly smaller size, about half the parameters of BERT large, making it a more efficient choice with comparable performance.

4. References

- [1] B. Liu and I. Lane, “Attention-based recurrent neural network models for joint intent detection and slot filling,” in *Proceedings of Interspeech 2016*, San Francisco, USA, Sep. 2016, pp. 685–689. [Online]. Available: https://www.isca-archive.org/interspeech2016/liu16c_interspeech.pdf
- [2] Q. Chen, Z. Zhuo, and W. Wang, “Bert for joint intent classification and slot filling,” *arXiv preprint arXiv:1902.10909*, 2019. [Online]. Available: <https://arxiv.org/abs/1902.10909>

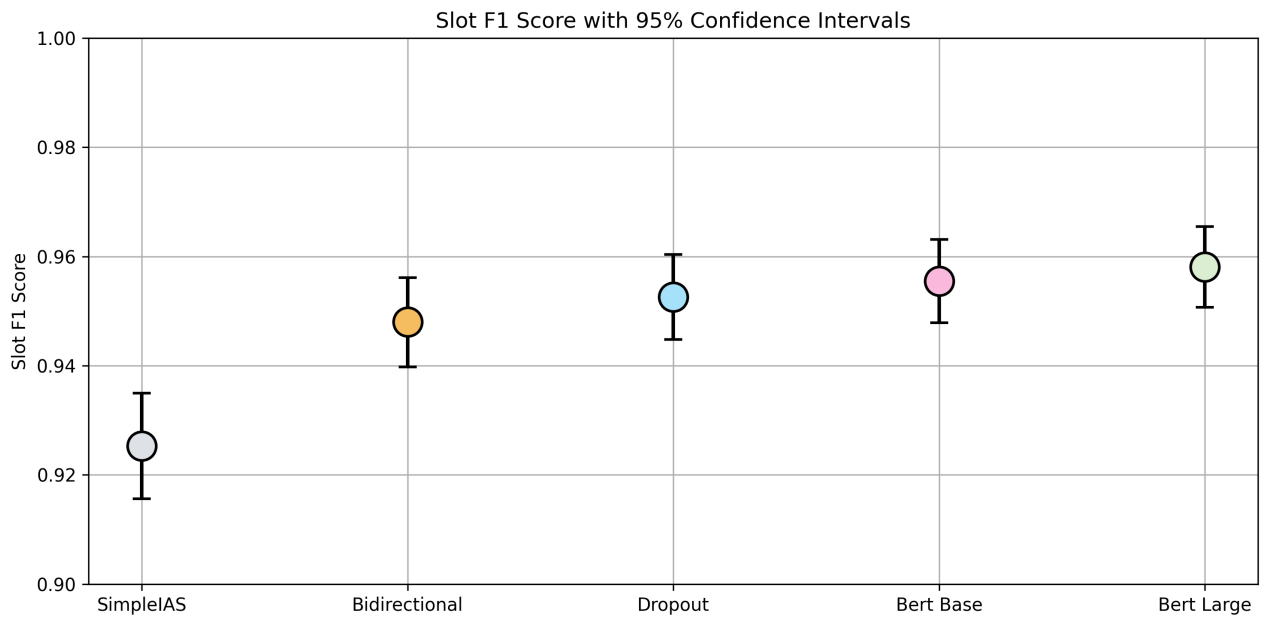


Figure 2

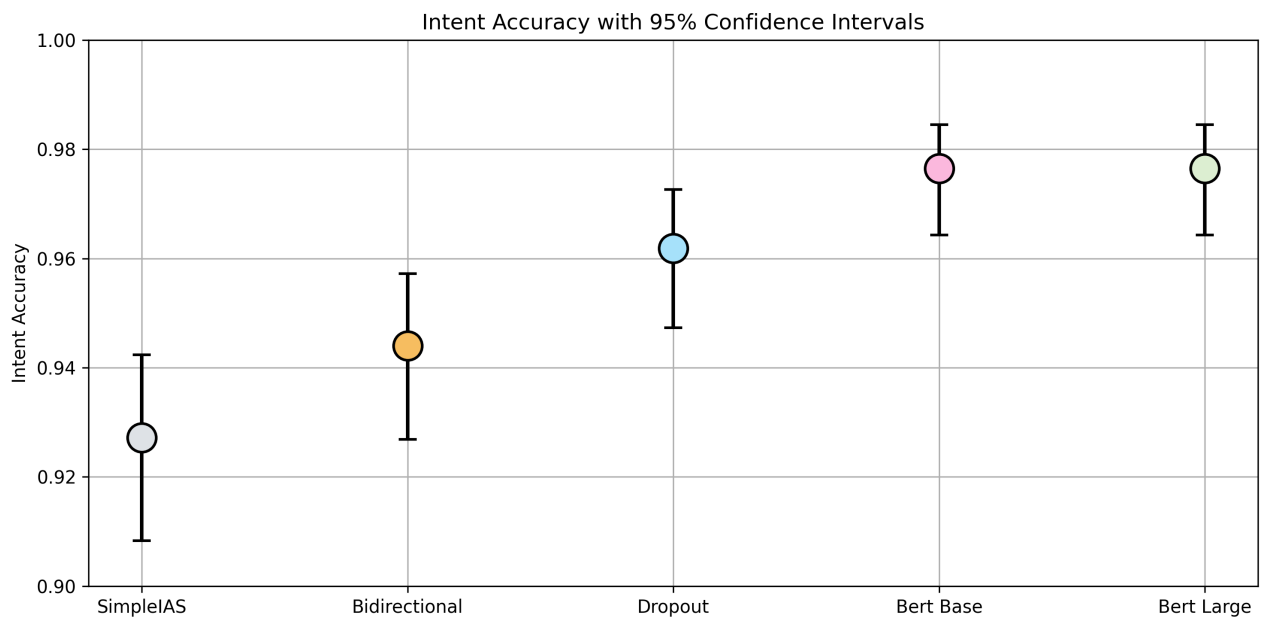


Figure 3